

Seedless One-Sided Partial Recovery in a Dyadic Power-Law Chung–Lu Model

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Abstract

We prove a seedless one-sided partial-recovery guarantee for a stylized dyadic power-law Chung–Lu graph with exponent $\beta \in (2, 3)$. Two observed graphs are obtained by independently retaining each parent edge with fixed probability $s \in (0, 1]$, after which one graph is relabeled by an unknown permutation. The main algorithm uses observed degrees to recover high-degree annuli, finds endogenous seeds by comparing coarse degree profiles, propagates those seeds down to logarithmic degree by one-hop witnesses, and then uses a layer-free global witness threshold at low degrees. With high probability, the algorithm returns $\Omega(n)$ correct pairs and no incorrect pairs. A cleaner revealed-layer theorem is stated and proved as the main scaffold. The result is model-specific and partial, but it operates at fixed correlation and in a graph with many constant-degree vertices, outside the high-minimum-degree, near-perfect-correlation hypotheses of Ding–Fei–Wang. [\[SJ: In progress; revising and generalizing with the help of AI.\]](#)

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1 Introduction

Correlated graph alignment asks one to match the vertex sets of two graphs generated from a common latent network. In homogeneous sparse models, seedless alignment is delicate: when degrees carry little information, the algorithm must exploit large combinatorial neighborhoods. In power-law graphs there is a different opportunity. The high-degree vertices, though few, have many independent local measurements. Their coarse degree profiles can identify a polynomial number of correct pairs without external seeds. Once these pairs have been found, standard one-hop witness propagation can carry the matching down through lower-degree layers.

This note isolates that mechanism in a deliberately simple dyadic Chung–Lu model. Vertices are partitioned into layers L_k . A vertex in L_k has weight 2^k , and $|L_k| \asymp n2^{-(\beta-1)k}$ for $\beta \in (2, 3)$. The two observed graphs are correlated subsamples of a common parent graph. The retention probability $s \in (0, 1]$ is fixed, so the result is not a near-perfect-correlation theorem.

The main message is the following.

Theorem 1.1 (Informal main theorem). *For every fixed $\beta \in (2, 3)$ and $s \in (0, 1]$, there is a polynomial-time seedless algorithm which, in the dyadic power-law model below, outputs a partial matching $\mathcal{S}$ such that*

$$\mathbb{P}[\mathcal{S} \text{ contains an incorrect pair}] \rightarrow 0, \quad \mathbb{P}[|\mathcal{S} \cap \{(v, \pi^*(v)) : v \in [n]\}| \geq cn] \rightarrow 1$$

for some constant $c = c(\beta, s) > 0$.

The proof has two layers. First, we prove an oracle-layer theorem in which the dyadic layers are revealed. This version exposes the main mechanism cleanly: profile-based seed discovery, full

one-hop propagation to height $\Theta(\log \log n)$, and threshold propagation to a constant-height layer containing $\Theta(n)$ vertices. Second, we remove the most artificial part of the oracle by observing that logarithmic-degree layers are recoverable from observed degrees, while the final low-degree stage can be implemented by a global witness threshold that does not use layer labels.

The result should be read as a model-specific complement to the degree-profile theorem of Ding et al. (2025), not as a dominance result. Ding–Fei–Wang prove exact seedless recovery for a far more general inhomogeneous model, but under high minimum expected degree and correlation tending to one. Here the model is much less general and the guarantee is only partial, but the algorithm works at fixed retention probability and with constant-degree vertices present.

2 Model and main results

2.1 Dyadic power-law Chung–Lu model

Fix $\beta \in (2, 3)$ and set

$$k_{\max} = \left\lfloor \frac{\log_2 n}{\beta - 1} \right\rfloor. \quad (1)$$

The vertex set $[n]$ is partitioned into layers $L_0, \dots, L_{k_{\max}}$. Let

$$C_1 = \left(\sum_{k=0}^{k_{\max}} 2^{-(\beta-1)k} \right)^{-1}, \quad (2)$$

and set $N_k = |L_k| = \lceil C_1 n 2^{-(\beta-1)k} \rceil$ for $k \geq 1$, with N_0 chosen so that the total number of vertices is n . A vertex in L_k receives weight $w_v = 2^k$, and

$$w_{\text{tot}} = \sum_{v \in [n]} w_v = \sum_{k=0}^{k_{\max}} 2^k N_k. \quad (3)$$

The parent graph G_{par} is the Chung–Lu graph in which, independently over pairs including self-loops,

$$\mathbb{P}[(u, v) \in E(G_{\text{par}})] = p_{k\ell} := 1 \wedge \frac{2^k 2^\ell}{w_{\text{tot}}}, \quad u \in L_k, v \in L_\ell. \quad (4)$$

Self-loops are allowed only to avoid irrelevant boundary notation; removing them changes none of the arguments.

Given G_{par} , construct G_1 and G_2 by retaining each parent edge independently in each child with probability $s \in (0, 1]$. Finally draw an unknown permutation π^* of $[n]$ and relabel G_2 by π^* . We write the resulting model as $\text{PLC}(n, \beta, s)$. In proofs, we identify the two vertex sets using the latent labels and take $\pi^* = \text{Id}$.

Definition 2.1 (One-sided partial recovery). *For a partial matching $\mathcal{S} \subseteq V(G_1) \times V(G_2)$, define*

$$\text{corr}(\mathcal{S}) = |\mathcal{S} \cap \{(v, \pi^*(v)) : v \in [n]\}|, \quad (5)$$

$$\text{err}(\mathcal{S}) = |\mathcal{S} \setminus \{(v, \pi^*(v)) : v \in [n]\}|. \quad (6)$$

An algorithm achieves one-sided partial recovery if, for some constant $c > 0$,

$$\mathbb{P}[\text{err}(\mathcal{S}) = 0] \rightarrow 1, \quad \mathbb{P}[\text{corr}(\mathcal{S}) \geq cn] \rightarrow 1. \quad (7)$$

The following elementary estimates are used throughout.

Lemma 2.2 (Layer calculus). *There are constants $0 < c < C < \infty$, depending only on β , such that the following hold for all large n .*

$$(i) \quad cn2^{-(\beta-1)k} \leq N_k \leq Cn2^{-(\beta-1)k} \text{ for every } 0 \leq k \leq k_{\max}.$$

$$(ii) \quad cn \leq w_{\text{tot}} \leq Cn.$$

$$(iii) \quad \text{Whenever truncation is inactive, } p_{k\ell} \asymp 2^{k+\ell}/n.$$

$$(iv) \quad \text{Whenever truncation is inactive,}$$

$$N_j p_{\ell j} \asymp 2^{\ell-(\beta-2)j}, \quad N_\ell p_{\ell j} \asymp 2^{j-(\beta-2)\ell}. \quad (8)$$

In particular,

$$N_j p_{j-1,j} \asymp 2^{(3-\beta)j}. \quad (9)$$

Proof. The first claim follows from the construction of N_k and the fact that $\sum_{k \geq 0} 2^{-(\beta-1)k} < \infty$. For the total weight,

$$w_{\text{tot}} \asymp n \sum_{k=0}^{k_{\max}} 2^k 2^{-(\beta-1)k} = n \sum_{k=0}^{k_{\max}} 2^{-(\beta-2)k} \asymp n, \quad (10)$$

because $\beta > 2$. The remaining displays follow by substituting $N_j \asymp n2^{-(\beta-1)j}$ and $p_{\ell j} \asymp 2^{\ell+j}/n$. $\square$

2.2 Main theorem without revealed layers

The headline result is the observed-degree version. The algorithm uses only G_1, G_2 , the known parameters β, s , and the dyadic form of the model.

Theorem 2.3 (Observed-degree seedless partial alignment). *Fix $\beta \in (2, 3)$ and $s \in (0, 1]$. In $\text{PLC}(n, \beta, s)$, suppose the latent layers are not revealed to the algorithm, but β and s are known. There is a polynomial-time algorithm, $\text{EMPIRICALANNULIALIGNMENT}$, whose output $\mathcal{S}$ satisfies, for some $c = c(\beta, s) > 0$,*

$$\mathbb{P}[\text{err}(\mathcal{S}) = 0] \rightarrow 1, \quad \mathbb{P}[\text{corr}(\mathcal{S}) \geq cn] \rightarrow 1. \quad (11)$$

The proof of Theorem 2.3 uses the following revealed-layer result as a scaffold.

Theorem 2.4 (Oracle-layer seedless partial alignment). *Fix $\beta \in (2, 3)$ and $s \in (0, 1]$. In $\text{PLC}(n, \beta, s)$, suppose the layer memberships $L_k^{(1)}, L_k^{(2)}$ are revealed. Then PROFILESEEDS , $\text{FULLWITNESSPROPAGATION}$, and $\text{THRESHOLDWITNESSPROPAGATION}$, run in sequence, achieve one-sided partial recovery in polynomial time.*

3 Algorithms

This section states the four routines used in the theorem. The pseudocode enforces one-to-one matching by always optimizing over unmatched candidates. On the good events in the proofs, the true counterpart is the unique optimizer whenever a match is made.

3.1 Profile seed discovery

For $v \in G_a$, let

$$X_\ell^{(a)}(v) = |N_{G_a}(v) \cap L_\ell^{(a)}|, \quad a \in \{1, 2\}. \quad (12)$$

For a target seed layer j and a profile layer ℓ , define

$$Z_\ell^{(n)}(v, u) = \frac{X_\ell^{(1)}(v) - X_\ell^{(2)}(u)}{\sqrt{N_\ell p_{\ell j}}}, \quad v \in L_j^{(1)}, u \in L_j^{(2)}. \quad (13)$$

For a profile index set $\mathcal{I}_{\text{prof}}$, set

$$T^{(n)}(v, u) = \frac{1}{|\mathcal{I}_{\text{prof}}|} \sum_{\ell \in \mathcal{I}_{\text{prof}}} |Z_\ell^{(n)}(v, u)|. \quad (14)$$

The seed statistic is smaller for true pairs because the two child counts share the same parent-neighborhood randomness. For $u \neq v$,

$$\mathbb{E} [T^{(n)}(v, u)] \approx \frac{2\sqrt{s}}{\sqrt{\pi}}, \quad \mathbb{E} [T^{(n)}(v, v)] \approx \frac{2\sqrt{s(1-s)}}{\sqrt{\pi}}, \quad (15)$$

so the asymptotic gap is

$$\Delta_s = \frac{2\sqrt{s}}{\sqrt{\pi}} (1 - \sqrt{1-s}) > 0. \quad (16)$$

Choose a sufficiently small constant $\alpha = \alpha(\beta, s) > 0$ and define

$$\mathcal{J}_{\text{seed}} = \left\{ j : k_{\max} - \left\lfloor \frac{\alpha}{\beta - 1} \log_2 n \right\rfloor \leq j \leq k_{\max} \right\}. \quad (17)$$

Algorithm 1 PROFILESEEDS

```

1: Input:  $G_1, G_2$ , revealed layers, parameters  $\beta, s$ , constants  $\alpha, c_0$ .
2: Output: a partial matching  $\mathcal{S}$ .
3:  $\mathcal{S} \leftarrow \emptyset$ .
4: for  $j \in \mathcal{J}_{\text{seed}}$  do
5:    $U_j \leftarrow L_j^{(2)}$ .
6:   for  $v \in L_j^{(1)}$  do
7:      $u^* \leftarrow \operatorname{argmin}_{u \in U_j} T^{(n)}(v, u)$ , computed using  $\mathcal{I}_{\text{prof}}$ .
8:      $\mathcal{S} \leftarrow \mathcal{S} \cup \{(v, u^*)\}$  and  $U_j \leftarrow U_j \setminus \{u^*\}$ .
9:   end for
10: end for
11: return  $\mathcal{S}$ .

```

Choose $c_0 > 0$ large and define the profile layers

$$\mathcal{I}_{\text{prof}} = \{0 \leq \ell \leq (\beta - 2)k_{\max} - c_0 \log_2 \log_2 n\}. \quad (18)$$

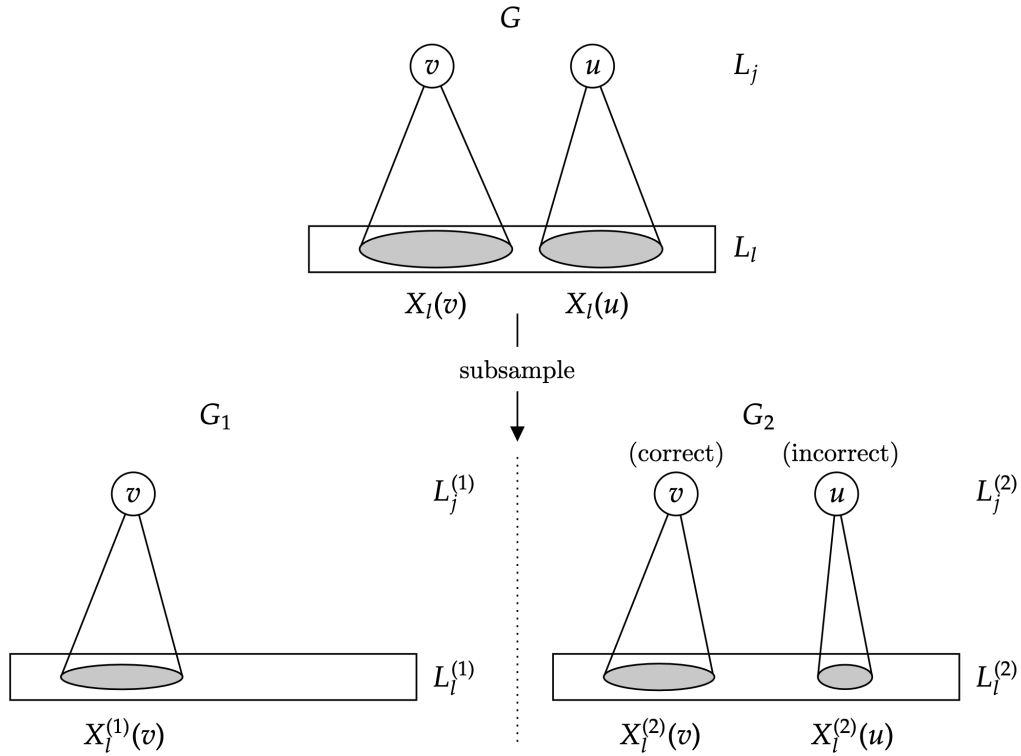


Figure 1: PROFILESEEDS. A true pair has more similar coarse degree profiles because the observed counts share the same parent-neighborhood randomness.

3.2 Full one-hop witness propagation

Suppose a layer L_j has already been fully and correctly matched. For $v \in L_\ell^{(1)}$ and $u \in L_\ell^{(2)}$, define the one-hop witness count

$$A_j^{(n)}(v, u) = |\{w \in L_j : (v, w) \in E(G_1), (u, w) \in E(G_2)\}|. \quad (19)$$

A true pair has mean $s^2 N_j p_{\ell_j}$, whereas a false pair has mean $s^2 N_j p_{\ell_j}^2$. Hence full recovery of layer L_ℓ is possible when $N_j p_{\ell_j} \gg \log n$ and $p_{\ell_j} \rightarrow 0$.

Let $j_{\min}$ be the smallest seed layer recovered by PROFILESEEDS. Define

$$\ell_{\max} = (\beta - 2)k_{\max} - c_0 \log_2 \log_2 n, \quad (20)$$

$$\ell_{\text{mid}} = (\beta - 2)j_{\min} + c_1 \log_2 \log_2 n, \quad (21)$$

$$\ell_{\min} = \frac{c_1}{3 - \beta} \log_2 \log_2 n, \quad (22)$$

where $c_1 > 1$ is fixed. The first propagation phase uses the matched seed layer $L_{j_{\min}}$ to match all layers from $\ell_{\max}$ down to ℓ_{mid} . The second phase descends one layer at a time from $\ell_{\text{mid}} - 1$ to $\ell_{\min}$.

Algorithm 2 FULLWITNESSPROPAGATION

- 1: **Input:** G_1, G_2 , revealed layers, a fully matched seed layer $L_{j_{\min}}$, parameters β, c_0, c_1 .
 - 2: **Output:** a set of matched pairs $\mathcal{S}$.
 - 3: $\mathcal{S} \leftarrow \emptyset$.
 - 4: **for** $\ell = \ell_{\max}, \ell_{\max} - 1, \dots, \ell_{\text{mid}}$ **do**
 - 5: Match $L_\ell^{(1)}$ to $L_\ell^{(2)}$ by assigning each $v \in L_\ell^{(1)}$ to an unmatched maximizer of $A_{j_{\min}}^{(n)}(v, u)$.
 - 6: Add these pairs to $\mathcal{S}$.
 - 7: **end for**
 - 8: **for** $\ell = \ell_{\text{mid}} - 1, \ell_{\text{mid}} - 2, \dots, \ell_{\min}$ **do**
 - 9: Match $L_\ell^{(1)}$ to $L_\ell^{(2)}$ by assigning each $v \in L_\ell^{(1)}$ to an unmatched maximizer of $A_{\ell+1}^{(n)}(v, u)$.
 - 10: Add these pairs to $\mathcal{S}$.
 - 11: **end for**
 - 12: **return** $\mathcal{S}$.
-

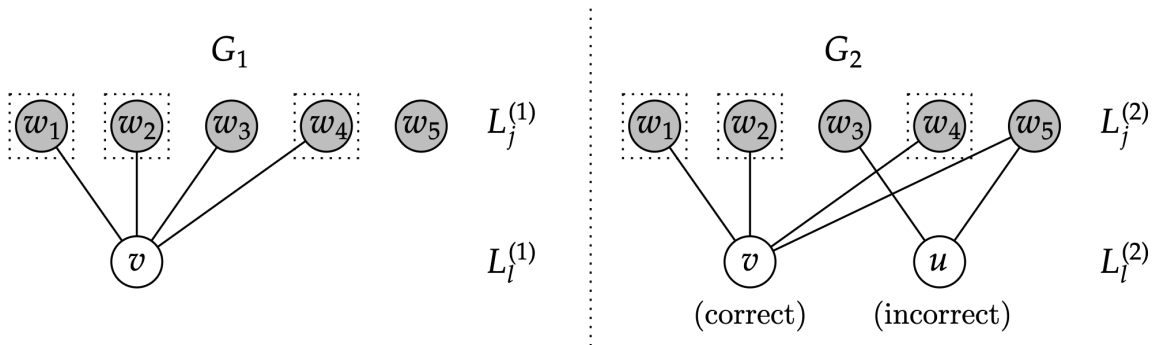


Figure 2: Full witness propagation. A true pair has many common one-hop witnesses in a previously matched layer, while false pairs have far fewer.

3.3 Threshold witness propagation

Below height $\Theta(\log \log n)$, full matching is no longer needed. If a subset $S_j \subseteq L_j \times L_j$ has been correctly matched, define

$$\tilde{A}_j^{(n)}(v, u) = |\{(w, w) \in S_j : (v, w) \in E(G_1), (u, w) \in E(G_2)\}|. \quad (23)$$

The threshold routine matches a pair only if its witness count is at least 3.

Let

$$g(x) = 1 - e^{-x} \left(1 + x + \frac{x^2}{2}\right), \quad (24)$$

the probability that a $\text{Poisson}(x)$ variable is at least 3. Choose a constant $j_{\text{thresh}} = j_{\text{thresh}}(\beta, s)$ large enough that

$$g(C_3 \lambda 2^{(3-\beta)j}) \geq \frac{1}{2} \quad \text{for all } j \geq j_{\text{thresh}} \text{ and all } \lambda \geq \frac{1}{2}, \quad (25)$$

where $C_3 > 0$ is the constant in Proposition 4.3 below.

Algorithm 3 THRESHOLDWITNESSPROPAGATION

- 1: **Input:** G_1, G_2 , revealed layers, a fully matched layer $L_{\ell_{\min}}$, threshold 3.
 - 2: **Output:** a partial matching $\mathcal{S}$.
 - 3: $S_{\ell_{\min}} \leftarrow \{(v, v) : v \in L_{\ell_{\min}}\}$ and $\mathcal{S} \leftarrow S_{\ell_{\min}}$.
 - 4: **for** $j = \ell_{\min}, \ell_{\min} - 1, \dots, j_{\text{thresh}} + 1$ **do**
 - 5: $S_{j-1} \leftarrow \emptyset$ and $U \leftarrow L_{j-1}^{(2)}$.
 - 6: **for** $v \in L_{j-1}^{(1)}$ **do**
 - 7: $u^* \leftarrow \operatorname{argmax}_{u \in U} \tilde{A}_j^{(n)}(v, u)$.
 - 8: **if** $\tilde{A}_j^{(n)}(v, u^*) \geq 3$ **then**
 - 9: $S_{j-1} \leftarrow S_{j-1} \cup \{(v, u^*)\}$ and $U \leftarrow U \setminus \{u^*\}$.
 - 10: **end if**
 - 11: **end for**
 - 12: $\mathcal{S} \leftarrow \mathcal{S} \cup S_{j-1}$.
 - 13: **end for**
 - 14: **return** $\mathcal{S}$.
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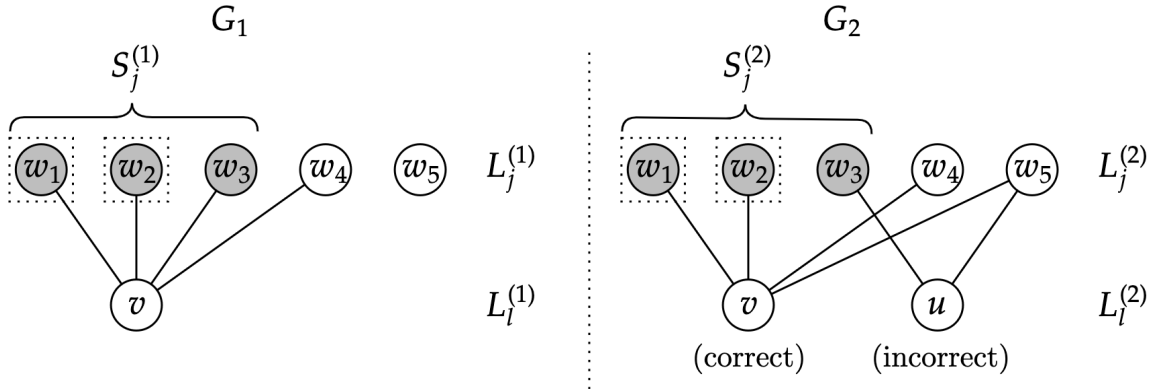


Figure 3: Threshold propagation. At constant height, the true pair only needs at least three witnesses; false pairs reach this threshold with vanishing probability after a union bound.

3.4 Observed-degree implementation

The observed-degree algorithm replaces latent layers by empirical annuli whenever expected degrees are logarithmic. Let $D_a(v)$ be the observed degree of v in G_a , and write

$$d_k = \mathbb{E} \left[\deg_{G_{\text{par}}}(v) \mid v \in L_k \right], \quad \mu_k = sd_k = \mathbb{E} [D_a(v) \mid v \in L_k]. \quad (26)$$

The quantities μ_k are known functions of (n, β, s) in the model. For a small fixed $\eta \in (0, 1/4)$ and for layers with $\mu_k \geq A \log n$, define

$$\widehat{L}_k^{(a)} = \{v \in V(G_a) : |D_a(v) - \mu_k| \leq \eta \mu_k\}. \quad (27)$$

For A sufficiently large, these annuli equal the latent layers simultaneously with high probability. The routines PROFILESEEDS and FULLWITNESSPROPAGATION are then run using $\widehat{L}_k^{(a)}$ in place of $L_k^{(a)}$ until the boundary $\ell_{\min} = \Theta(\log \log n)$.

At the final stage, degrees are too noisy to recover constant-height layers. Instead, the algorithm freezes all vertices placed into recovered high-degree annuli and runs a global threshold-3 witness rule over the remaining candidate pool. This is EMPIRICALANNULIALIGNMENT.

Algorithm 4 EMPIRICALANNULIALIGNMENT

- 1: **Input:** G_1, G_2 , known β, s , constants $A, \eta, c_0, c_1, \alpha$.
 - 2: Construct empirical annuli $\widehat{L}_k^{(1)}, \widehat{L}_k^{(2)}$ for all k with $\mu_k \geq A \log n$.
 - 3: Run PROFILESEEDS using empirical annuli in place of latent layers.
 - 4: Run FULLWITNESSPROPAGATION using empirical annuli down to $\ell_{\min}$.
 - 5: Let U_1, U_2 be the unmatched vertices not assigned to recovered annuli above the boundary.
 - 6: Run the global low-degree threshold-3 witness routine over $U_1 \times U_2$, using the cumulative already matched set as witnesses.
 - 7: **return** all pairs matched in the three stages.
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4 Proof architecture

The proof of the oracle theorem is the concatenation of three propositions.

Proposition 4.1 (Profile seeds). *There exists $\alpha = \alpha(\beta, s) > 0$ such that PROFILESEEDS fully and correctly matches every layer L_j with $j \in \mathcal{J}_{\text{seed}}$ with probability $1 - o(1)$. In particular, it produces at least n^α correct seeds and no incorrect seed with probability $1 - o(1)$.*

Proposition 4.2 (Full witness wave). *Suppose L_j has been fully and correctly matched. Let ℓ be a lower-degree layer such that truncation is inactive, $p_{\ell j} \rightarrow 0$, and*

$$N_j p_{\ell j} \geq C \log n \quad (28)$$

for a sufficiently large constant $C = C(\beta, s)$. Then matching $L_\ell^{(1)}$ to $L_\ell^{(2)}$ by maximizing $A_j^{(n)}(v, u)$ recovers all of L_ℓ correctly with probability $1 - n^{-10}$.

Proposition 4.3 (Threshold witness wave). *There is a constant $C_3 = C_3(\beta, s) > 0$ with the following property. Suppose S_j contains no false pair and contains at least a $\lambda_j \geq 1/2$ fraction of the true pairs in L_j . Then one threshold-3 wave from L_j to L_{j-1} produces no false pair with probability $1 - o((\log n)^{-1})$ and matches at least*

$$(1 - n^{-1/3}) g(C_3 \lambda_j 2^{(3-\beta)j}) \quad (29)$$

of the true pairs in L_{j-1} with probability $1 - o((\log n)^{-1})$.

Proof of Theorem 2.4. By Proposition 4.1, the seed stage fully and correctly matches every layer in $\mathcal{J}_{\text{seed}}$. Let $j_{\min}$ be the lowest of these seed layers. We apply Proposition 4.2 in the two regimes encoded by FULLWITNESSPROPAGATION. For the first regime, with witness layer $j_{\min}$,

$$N_{j_{\min}} p_{\ell, j_{\min}} \asymp 2^{\ell - (\beta-2)j_{\min}} \geq (\log n)^{c_1} \quad (30)$$

for all $\ell \in [\ell_{\text{mid}}, \ell_{\text{max}}]$. For the adjacent regime,

$$N_{\ell+1} p_{\ell, \ell+1} \asymp 2^{(3-\beta)\ell} \geq (\log n)^{c_1} \quad (31)$$

for all $\ell \geq \ell_{\min}$. A union bound over $O(\log n)$ layers shows that every layer down to $\ell_{\min}$ is fully and correctly matched with probability $1 - o(1)$.

Starting from the fully matched layer $L_{\ell_{\min}}$, set $\lambda_{\ell_{\min}} = 1$ and apply Proposition 4.3 downward. Suppose inductively that S_j contains no false pair and contains at least a $\lambda_j \geq 1/2$ fraction of L_j . By the choice of j_{thresh} , for every $j \geq j_{\text{thresh}} + 1$,

$$g(C_3 \lambda_j 2^{(3-\beta)j}) \geq \frac{1}{2}. \quad (32)$$

The proposition therefore gives

$$\lambda_{j-1} \geq (1 - n^{-1/3}) g(C_3 \lambda_j 2^{(3-\beta)j}) \geq \frac{1}{2} - o(1). \quad (33)$$

Increasing j_{thresh} by a fixed amount, if necessary, absorbs the $o(1)$ term and keeps the invariant $\lambda_{j-1} \geq 1/2$ for all large n . There are only $O(\log \log n)$ threshold waves, and each has false-pair probability $o((\log n)^{-1})$, so a union bound gives no false pair throughout the threshold stage with probability $1 - o(1)$. The layer $L_{j_{\text{thresh}}}$ has size $N_{j_{\text{thresh}}} = \Theta(n)$ because j_{thresh} is a constant depending only on β, s . Thus at least a constant fraction of n vertices are correctly matched and $\text{err}(\mathcal{S}) = 0$ with probability tending to one. $\square$

5 Removing the layer oracle

The revealed-layer theorem uses exact layer labels in two places: to form the high-degree profile bins and to restrict the witness propagation to one target layer at a time. Both uses can be relaxed.

Lemma 5.1 (Empirical recovery of logarithmic-degree annuli). *Fix $\eta \in (0, 1/4)$ and $M > 0$. There exists $A = A(\beta, s, \eta, M)$ such that, with probability at least $1 - n^{-M}$, the following holds simultaneously for both observed graphs and all layers satisfying $\mu_k \geq A \log n$:*

$$\widehat{L}_k^{(a)} = L_k^{(a)}, \quad a \in \{1, 2\}. \quad (34)$$

Moreover $d_k = (1 + o(1))2^k$ uniformly over $0 \leq k \leq k_{\max}$, so the recoverable-annulus condition is equivalent to $s2^k \geq (1 + o(1))A \log n$.

Proof. Without truncation in the Chung–Lu probabilities, the expected parent degree of a vertex in L_k would be exactly 2^k . Therefore

$$0 \leq 2^k - d_k = \sum_{\ell=0}^{k_{\max}} N_\ell \left(\frac{2^k 2^\ell}{w_{\text{tot}}} - 1 \right)_+. \quad (35)$$

Since $w_{\text{tot}} = \Theta(n)$, the summands are nonzero only for $\ell \geq L(k) := \lceil \log_2(cn) - k \rceil$ for a constant $c > 0$. Thus

$$2^k - d_k \leq C2^k \sum_{\ell \geq L(k)} 2^{-(\beta-2)\ell} \leq C'2^k 2^{-(\beta-2)L(k)}. \quad (36)$$

Dividing by 2^k and using $k \leq k_{\max}$ gives

$$0 \leq 1 - \frac{d_k}{2^k} \leq C''n^{-(\beta-2)}2^{(\beta-2)k} \leq C''n^{-(\beta-2)^2/(\beta-1)} = o(1), \quad (37)$$

uniformly in k . Hence adjacent means μ_k are eventually separated by a factor $2 + o(1)$, so the intervals $[(1 - \eta)\mu_k, (1 + \eta)\mu_k]$ are pairwise disjoint for all sufficiently large n .

For $v \in L_k$, the observed degree $D_a(v)$ is a sum of independent Bernoulli variables with mean μ_k . If $\mu_k \geq A \log n$, Chernoff's bound gives

$$\mathbb{P}[|D_a(v) - \mu_k| > \eta\mu_k] \leq 2 \exp(-c_\eta \mu_k) \leq 2n^{-M-4} \quad (38)$$

for A sufficiently large. A union bound over both graphs and all vertices in recoverable layers puts every recoverable-layer vertex in its own annulus with probability at least $1 - O(n^{-M-3})$. On this event, a vertex from one recoverable layer cannot also lie in another recoverable annulus because the annuli are pairwise disjoint.

It remains to exclude vertices from nonrecoverable layers. Let k satisfy $\mu_k < A \log n$, and let r be recoverable. For all large n , recoverability and the adjacent-ratio estimate imply $r > k$ and

$$\frac{(1 - \eta)\mu_r}{\mu_k} \geq (1 - \eta)(2 - o(1)) > 1 + \delta_\eta \quad (39)$$

for some $\delta_\eta > 0$. Moreover $(1 - \eta)\mu_r \geq (1 - \eta)A \log n$. The Chernoff upper-tail bound in relative-entropy form says that, for $t > \mu_k$,

$$\mathbb{P}[D_a(v) \geq t] \leq \exp \left[-t \left\{ \log \left(\frac{t}{\mu_k} \right) - 1 + \frac{\mu_k}{t} \right\} \right]. \quad (40)$$

Since $t/\mu_k \geq 1 + \delta_\eta$ at $t = (1 - \eta)\mu_r$, the quantity in braces is bounded below by a positive constant depending only on η . Therefore

$$\mathbb{P}[D_a(v) \in [(1 - \eta)\mu_r, (1 + \eta)\mu_r]] \leq \mathbb{P}[D_a(v) \geq (1 - \eta)\mu_r] \leq n^{-M-4} \quad (41)$$

for A large enough, uniformly over all nonrecoverable k and recoverable r . A union bound over two graphs, n vertices, and $O(\log n)$ recoverable annuli shows that no nonrecoverable-layer vertex enters any recoverable annulus with probability at least $1 - O(n^{-M-2})$. Combining the two events proves the claim, after increasing A once more if necessary. $\square$

After the empirical-annulus stage, the algorithm has matched down to height $\ell_{\min} = \Theta(\log \log n)$. It then freezes all vertices assigned to recovered high-degree annuli. The remaining candidate pools contain only vertices of latent height at most $J = \ell_{\min} - 1 = O(\log \log n)$ on the event of Lemma 5.1. The final propagation step uses no latent labels.

For a cumulative matched set S and candidate pair (v, u) , define the global witness score

$$W_S(v, u) = |\{(w, w) \in S : (v, w) \in E(G_1), (u, w) \in E(G_2)\}|. \quad (42)$$

The global low-degree wave processes unmatched vertices $v \in U_1$, chooses an unmatched $u \in U_2$ maximizing $W_S(v, u)$, and accepts the pair only if $W_S(v, u) \geq 3$.

Proposition 5.2 (Global low-degree wave without layer labels). *Let $J = O(\log \log n)$. Suppose the candidate pools U_1, U_2 contain only vertices of latent height at most J . Suppose every vertex of L_{j-1} is either already correctly matched in the cumulative set S or is present in the candidate pools in both graphs. Suppose also that S contains no incorrect pairs and, for some $j \leq J$, contains at least a $\lambda_j \geq 1/2$ fraction of the true pairs in L_j . The set S may also contain correctly matched pairs from any layers between j and J . Then one global threshold-3 wave over $U_1 \times U_2$ produces no incorrect match with probability $1 - o((\log n)^{-1})$ and, after adding the accepted pairs to S , the cumulative matching contains at least*

$$(1 - n^{-1/3}) g(C_3 \lambda_j 2^{(3-\beta)j}) \quad (43)$$

of the true pairs in L_{j-1} with probability $1 - o((\log n)^{-1})$.

Proof. First consider true pairs in L_{j-1} . Let $S_j \subseteq S$ denote the correctly matched subset from layer L_j , so $|S_j| \geq \lambda_j N_j$. Let $R_{j-1} \subseteq L_{j-1}$ be the vertices not already matched in S ; the vertices in $L_{j-1} \setminus R_{j-1}$ already count toward the desired cumulative total. For $v \in R_{j-1}$, the score $W_S(v, v)$ is at least the number of witnesses coming only from S_j . Conditional on S_j , that restricted count is

$$\text{Bin}(|S_j|, s^2 p_{j-1,j}), \quad (44)$$

because the edges from v to distinct witnesses in L_j are independent in the parent graph and are retained independently in the two children. Therefore the same binomial calculation used in Proposition 4.3 gives

$$q_j := \mathbb{P}[W_S(v, v) \geq 3 \mid S_j] \geq (1 - o(1)) g(C_3 \lambda_j 2^{(3-\beta)j}), \quad (45)$$

uniformly over $v \in R_{j-1}$. For distinct $v \in R_{j-1}$ these restricted witness counts use disjoint edge sets, so the number of vertices in R_{j-1} whose true score reaches threshold stochastically dominates $\text{Bin}(|R_{j-1}|, q_j)$. Chernoff's lower-tail bound implies that, with probability $1 - o((\log n)^{-1})$, at least $(1 - n^{-1/3})q_j|R_{j-1}|$ such remaining vertices have true score at least 3. On the no-false-pair event proved below, each of those vertices is accepted with its true counterpart. If a fraction a of L_{j-1} was already matched before the wave, the final matched fraction in L_{j-1} is at least

$$a + (1 - a)(1 - n^{-1/3})q_j \geq (1 - n^{-1/3})q_j, \quad (46)$$

which gives the displayed lower bound after absorbing the $o(1)$ term.

It remains to control false pairs when the witness set is cumulative. We use a domination that avoids any adaptivity issue in the construction of S . For a mismatched pair $(v, u) \in U_1 \times U_2$, define

$$\overline{W}_J(v, u) = |\{w : h(w) \leq J, w \notin \{v, u\}, (v, w) \in E(G_1), (u, w) \in E(G_2)\}|, \quad (47)$$

where $h(w)$ is the latent layer index. Since S contains only already matched vertices and the candidate pools contain only unmatched vertices, every witness counted by $W_S(v, u)$ is also counted by $\overline{W}_J(v, u)$. Thus $W_S(v, u) \leq \overline{W}_J(v, u)$ deterministically.

Both vertices in the mismatched pair have latent height at most J . If $w \in L_t$ with $t \leq J$, then the two child edges needed for w to be a false witness have probability at most

$$C \left(\frac{2^{J+t}}{n} \right)^2 = C \frac{2^{2J+2t}}{n^2}, \quad (48)$$

using truncation inactivity for $J = O(\log \log n)$. The indicators in $\overline{W}_J(v, u)$ are independent over $w \notin \{v, u\}$, and their total mean satisfies

$$\nu_J \leq C \frac{2^{2J}}{n^2} \sum_{t \leq J} N_t 2^{2t} \quad (49)$$

$$\leq C' \frac{2^{2J}}{n^2} \cdot n 2^{(3-\beta)J} = n^{-1+o(1)}, \quad (50)$$

where we used Lemma 2.2 and $J = O(\log \log n)$. For a sum of independent Bernoulli variables with total mean ν_J ,

$$\mathbb{P} [\overline{W}_J(v, u) \geq 3] \leq \mathbb{E} \left[\binom{\overline{W}_J(v, u)}{3} \right] \leq \frac{\nu_J^3}{6} = n^{-3+o(1)}. \quad (51)$$

Therefore $\mathbb{P} [W_S(v, u) \geq 3] \leq n^{-3+o(1)}$ for every mismatched pair. A union bound over at most n^2 mismatched pairs gives no false threshold crossing with probability $1 - n^{-1+o(1)} = 1 - o((\log n)^{-1})$ in the current wave. $\square$

Proof of Theorem 2.3. Choose A in Lemma 5.1 large enough to dominate all union bounds. On the annulus-recovery event, every empirical annulus used by PROFILESEEDS and FULLWITNESSPROPAGATION equals the corresponding latent layer. Therefore the statistics $T^{(n)}(v, u)$ and $A_j^{(n)}(v, u)$ computed by the observed-degree algorithm are exactly the statistics

analyzed in Propositions 4.1 and 4.2 for every high-degree layer used before the boundary. The lowest layer used in the full-propagation stage has expected observed degree

$$s2^{\ell_{\min}+O(1)} = (\log n)^{c_1/(3-\beta)+o(1)} \gg \log n, \quad (52)$$

so all layers used in these two stages are recoverable annuli after increasing A by a fixed constant. Hence the seed and full-propagation conclusions hold down to $L_{\ell_{\min}}$ with probability $1 - o(1)$.

For the low-degree stage, remove from the candidate pools all vertices assigned to recovered annuli above the boundary. On the same event, the remaining vertices have latent height at most $J = \ell_{\min} - 1 = O(\log \log n)$ and include the lower layers to be recovered. Starting from the fully matched boundary layer, apply Proposition 5.2 for $j = \ell_{\min}, \ell_{\min} - 1, \dots, j_{\text{thresh}} + 1$. The induction from the oracle proof applies because each wave uses only the already matched cumulative set as witnesses, the proposition supplies the same lower bound for the next layer, and the proposition also rules out false matches anywhere in the global candidate pool. A union bound over $O(\log \log n)$ waves gives no false match with probability $1 - o(1)$. Since $N_{j_{\text{thresh}}} = \Theta(n)$, the algorithm outputs $\Omega(n)$ correct pairs and no incorrect pairs with probability tending to one. $\square$

6 Relation to prior work

Most practical and theoretical alignment algorithms use seeds: a small set of revealed correct pairs. Percolation-style seeded algorithms grow such a matching by counting one-hop witnesses, starting with the work of Yartseva and Grossglauser (2013) and refinements such as NOISYSEEDS and EXPANDWHILESTUCK in Kazemi et al. (2015). In power-law graphs, hubs make this strategy especially natural. Korula and Lattanzi’s USERMATCHING algorithm for preferential attachment graphs uses decreasing degree thresholds and requires many seeds Korula and Lattanzi (2013). Bringmann, Friedrich, and Krohmer analyze seeded alignment for a Chung–Lu power-law model and obtain constant-fraction recovery from the highest-degree seeds Bringmann et al. (2014). Chiasserini, Garetto, and Leonardi’s DEGREE-DRIVEN MATCHING slices vertices by degree and propagates matches downward Chiasserini et al. (2016). Yu, Xu, and Lin show that using D -hop witnesses can reduce the number of required seeds dramatically in power-law graphs Yu et al. (2021).

The present note keeps the one-hop propagation philosophy but removes external seeds in this stylized model. The high-degree layers generate their own seeds via coarse degree profiles, and those seeds then percolate downward.

The closest seedless comparison is Ding–Fei–Wang Ding et al. (2025), who prove exact recovery for broad inhomogeneous random graphs using continuum degree profiles, extending the Erdős–Rényi degree-profile method of Ding et al. (2021). Their theorem is much more general and exact, but it assumes minimum average degree at least a sufficiently large multiple of $(\log n)^2$ and edge correlations tending to one at rate $1 - O((\log n)^{-2})$. Our theorem is weaker in model generality and recovery objective, but it applies to a full power-law graph with constant-degree layers and fixed retention probability.

Thus the results are best viewed as incomparable. For the full dyadic graph here, the minimum expected degree is constant, so Ding–Fei–Wang does not directly apply to all vertices. In a high-

degree, near-perfect-correlation subregime satisfying their hypotheses, their theorem is stronger: it is exact, robust to much more general inhomogeneity, and parameter-free at the algorithmic level.

A Concentration tools for profile statistics

We collect the concentration facts used in Proposition 4.1. They are stated in the form needed for the seed proof.

Lemma A.1 (Binomial absolute-difference asymptotics). *Let Y_m, Y'_m be independent $\text{Bin}(m, q_m)$ variables with $mq_m \rightarrow \infty$ and $q_m \rightarrow 0$. Then*

$$\mathbb{E} \left[\left| \frac{Y_m - Y'_m}{\sqrt{mq_m}} \right| \right] \rightarrow \frac{2}{\sqrt{\pi}}. \quad (53)$$

If $B_m \sim \text{Bin}(m, q_m)$ and, conditionally on B_m , Y_m, Y'_m are independent $\text{Bin}(B_m, s)$ variables, then

$$\mathbb{E} \left[\left| \frac{Y_m - Y'_m}{\sqrt{mq_m}} \right| \right] \rightarrow \frac{2\sqrt{s(1-s)}}{\sqrt{\pi}}. \quad (54)$$

For independent child counts with marginal parameter sq_m , the corresponding limit is $2\sqrt{s}/\sqrt{\pi}$.

Proof. We use the triangular-array central limit theorem and then justify passage to absolute first moments by uniform integrability.

First let Y_m, Y'_m be independent $\text{Bin}(m, q_m)$ variables. Write

$$Y_m - Y'_m = \sum_{i=1}^m \xi_{m,i}, \quad \xi_{m,i} = \xi_{m,i}^{(1)} - \xi_{m,i}^{(2)}, \quad (55)$$

where $\xi_{m,i}^{(1)}, \xi_{m,i}^{(2)}$ are independent $\text{Bern}(q_m)$ variables. Then

$$\mathbb{E} [\xi_{m,i}] = 0, \quad \text{Var}(\xi_{m,i}) = 2q_m(1 - q_m), \quad |\xi_{m,i}| \leq 1. \quad (56)$$

Thus

$$\text{Var} \left(\frac{Y_m - Y'_m}{\sqrt{mq_m}} \right) = 2(1 - q_m) \rightarrow 2. \quad (57)$$

The Lindeberg condition holds because every normalized summand is bounded by $(mq_m)^{-1/2} \rightarrow 0$. Hence

$$\frac{Y_m - Y'_m}{\sqrt{mq_m}} \Rightarrow N(0, 2). \quad (58)$$

The independent-child case with marginal parameter sq_m is identical, with variance

$$\frac{m \cdot 2sq_m(1 - sq_m)}{mq_m} \rightarrow 2s, \quad (59)$$

so the limit is $N(0, 2s)$.

For the shared-parent case, write the contribution of potential neighbor i as

$$\zeta_{m,i} = B_{m,i}(R_{m,i}^{(1)} - R_{m,i}^{(2)}), \quad (60)$$

where $B_{m,i} \sim \text{Bern}(q_m)$ is the parent-edge indicator and $R_{m,i}^{(1)}, R_{m,i}^{(2)} \sim \text{Bern}(s)$ are independent child-retention indicators, independent of $B_{m,i}$. Then

$$Y_m - Y'_m = \sum_{i=1}^m \zeta_{m,i}, \quad \mathbb{E}[\zeta_{m,i}] = 0, \quad \text{Var}(\zeta_{m,i}) = q_m \mathbb{E}[(R_{m,i}^{(1)} - R_{m,i}^{(2)})^2] = 2s(1-s)q_m. \quad (61)$$

Again $|\zeta_{m,i}| \leq 1$, so Lindeberg follows from $mq_m \rightarrow \infty$. Therefore

$$\frac{Y_m - Y'_m}{\sqrt{mq_m}} \Rightarrow N(0, 2s(1-s)), \quad (62)$$

with the interpretation of a degenerate normal when $s = 1$.

It remains only to pass from weak convergence to convergence of expected absolute values. In all three cases the normalized variables have uniformly bounded second moments: for a constant $C_s < \infty$,

$$\sup_m \mathbb{E} \left[\left(\frac{Y_m - Y'_m}{\sqrt{mq_m}} \right)^2 \right] \leq C_s. \quad (63)$$

Consequently, for $K > 0$,

$$\mathbb{E} \left[\left| \frac{Y_m - Y'_m}{\sqrt{mq_m}} \right| \mathbf{1} \left\{ \left| \frac{Y_m - Y'_m}{\sqrt{mq_m}} \right| > K \right\} \right] \leq \frac{C_s}{K}, \quad (64)$$

uniformly in m . The absolute values are uniformly integrable. Therefore convergence in distribution implies convergence of the absolute first moments. Since $\mathbb{E}[|N(0, \sigma^2)|] = \sigma\sqrt{2/\pi}$, the three limits are respectively

$$\frac{2}{\sqrt{\pi}}, \quad \frac{2\sqrt{s(1-s)}}{\sqrt{\pi}}, \quad \frac{2\sqrt{s}}{\sqrt{\pi}}. \quad (65)$$

□

Lemma A.2 (Subexponential concentration of profile averages). *For the profile index set $\mathcal{I}_{\text{prof}}$ and seed layers $j \in \mathcal{J}_{\text{seed}}$, the random variables $|Z_\ell^{(n)}(v, u)|$, $\ell \in \mathcal{I}_{\text{prof}}$, are independent over ℓ and have uniformly bounded exponential moments in a neighborhood of zero. Consequently, for every fixed $\delta > 0$ there are constants $c, C > 0$ such that, uniformly over $j \in \mathcal{J}_{\text{seed}}$ and $v, u \in L_j$,*

$$\mathbb{P} \left[|T^{(n)}(v, u) - \mathbb{E}[T^{(n)}(v, u)]| > \delta \right] \leq C \exp(-c|\mathcal{I}_{\text{prof}}|). \quad (66)$$

Proof. Fix $j \in \mathcal{J}_{\text{seed}}$ and $\ell \in \mathcal{I}_{\text{prof}}$, and put

$$r_{\ell j} = N_\ell p_{\ell j}. \quad (67)$$

By Lemma 2.2 and the definitions of $\mathcal{J}_{\text{seed}}$ and $\mathcal{I}_{\text{prof}}$, after choosing the seed-window constant $\alpha > 0$ sufficiently small,

$$r_{\ell j} \geq n^{c'_0} \quad (68)$$

for some $c'_0 = c'_0(\beta, \alpha) > 0$, uniformly over all admissible (ℓ, j) . Also

$$p_{\ell j} \leq C \frac{2^{\ell+j}}{n} \leq C(\log n)^{-c_0}, \quad (69)$$

so $p_{\ell j} \rightarrow 0$ uniformly.

For either a true pair or a false pair, the numerator of $Z_\ell^{(n)}(v, u)$ can be written as

$$\sum_{i=1}^{N_\ell} \xi_i, \quad (70)$$

where the ξ_i are independent, mean-zero random variables satisfying

$$|\xi_i| \leq 1, \quad \sum_{i=1}^{N_\ell} \text{Var}(\xi_i) \leq C r_{\ell j}. \quad (71)$$

For a false pair, ξ_i is the difference of two independent $\text{Bern}(sp_{\ell j})$ indicators. For a true pair, $\xi_i = B_i(R_i^{(1)} - R_i^{(2)})$, with $B_i \sim \text{Bern}(p_{\ell j})$ and independent $R_i^{(1)}, R_i^{(2)} \sim \text{Bern}(s)$. We use the following elementary Bernstein mgf bound: if X_i are independent, mean-zero, $|X_i| \leq 1$, and $V = \sum_i \mathbb{E}[X_i^2]$, then for $|t| < 3$,

$$\mathbb{E} \left[\exp \left(t \sum_i X_i \right) \right] \leq \exp \left(\frac{t^2 V}{2(1 - |t|/3)} \right). \quad (72)$$

Taking $t = \theta / \sqrt{r_{\ell j}}$ and using $r_{\ell j} \rightarrow \infty$ gives, for every fixed θ and all large n ,

$$\begin{aligned} \mathbb{E} \left[\exp \left(\theta Z_\ell^{(n)}(v, u) \right) \right] &= \mathbb{E} \left[\exp \left(\frac{\theta}{\sqrt{r_{\ell j}}} \sum_{i=1}^{N_\ell} \xi_i \right) \right] \\ &\leq \exp(C\theta^2), \end{aligned} \quad (73)$$

uniformly over admissible ℓ, j, v, u . Applying the same bound to $-Z_\ell^{(n)}(v, u)$ yields

$$\mathbb{E} \left[\exp \left(\theta |Z_\ell^{(n)}(v, u)| \right) \right] \leq 2 \exp(C\theta^2) \quad (74)$$

for every fixed θ and all large n .

Let

$$Y_\ell = |Z_\ell^{(n)}(v, u)| - \mathbb{E} \left[|Z_\ell^{(n)}(v, u)| \right]. \quad (75)$$

The preceding display implies that, for some $\theta_* > 0$ and $K < \infty$,

$$\mathbb{E} \left[e^{\theta_* |Y_\ell|} \right] \leq K \quad (76)$$

uniformly over all admissible indices. Since $\mathbb{E}[Y_\ell] = 0$ and

$$|e^x - 1 - x| \leq \frac{x^2}{2} e^{|x|}, \quad (77)$$

we have, for $|\theta| \leq \theta_*/2$,

$$\mathbb{E}[e^{\theta Y_\ell}] \leq 1 + \frac{\theta^2}{2} \mathbb{E}[Y_\ell^2 e^{|\theta| |Y_\ell|}] \leq 1 + C_1 \theta^2 \leq \exp(C_1 \theta^2), \quad (78)$$

where the middle expectation is uniformly bounded by the exponential-moment bound. Thus there are constants $\theta_0, c_1 < \infty$ such that

$$\mathbb{E}[e^{\theta Y_\ell}] \leq \exp(c_1 \theta^2), \quad |\theta| \leq \theta_0, \quad (79)$$

uniformly over all admissible indices. The variables Y_ℓ , $\ell \in \mathcal{I}_{\text{prof}}$, are independent because the profile layers are disjoint and therefore use disjoint sets of potential edges. For any fixed $\delta > 0$, Chernoff's bound gives

$$\mathbb{P}\left[\frac{1}{|\mathcal{I}_{\text{prof}}|} \sum_{\ell \in \mathcal{I}_{\text{prof}}} Y_\ell > \delta\right] \leq \exp[-\theta \delta |\mathcal{I}_{\text{prof}}| + c_1 \theta^2 |\mathcal{I}_{\text{prof}}|] \quad (80)$$

for every $0 < \theta \leq \theta_0$. Taking $\theta = \min\{\theta_0, \delta/(2c_1)\}$ gives the upper-tail bound $\exp(-c|\mathcal{I}_{\text{prof}}|)$. The lower tail is identical with $-Y_\ell$. This proves the displayed concentration inequality, after adjusting the prefactor C . $\square$

B Proof of profile seed discovery

Proof of Proposition 4.1. Fix $j \in \mathcal{J}_{\text{seed}}$ and $v \in L_j^{(1)}$. For the true pair (v, v) , Lemma A.1 gives

$$\mathbb{E}[T^{(n)}(v, v)] = \frac{2\sqrt{s(1-s)}}{\sqrt{\pi}} + o(1). \quad (81)$$

For any false pair (v, u) with $u \neq v$ in $L_j^{(2)}$,

$$\mathbb{E}[T^{(n)}(v, u)] = \frac{2\sqrt{s}}{\sqrt{\pi}} + o(1). \quad (82)$$

The gap is $\Delta_s + o(1)$, where $\Delta_s > 0$. Taking $\delta = \Delta_s/4$ in Lemma A.2, we get

$$\mathbb{P}[T^{(n)}(v, v) \geq T^{(n)}(v, u)] \leq C \exp(-c|\mathcal{I}_{\text{prof}}|) \quad (83)$$

for every fixed false u , with $c = c(\beta, s) > 0$ after shrinking the seed window if necessary.

The total number of candidate comparisons over all layers in $\mathcal{J}_{\text{seed}}$ is at most $n^{2\alpha+o(1)}$. Since $|\mathcal{I}_{\text{prof}}| = \Theta(\log n)$, choose $\alpha = \alpha(\beta, s) > 0$ small enough that the union bound over all seed-layer comparisons is $o(1)$. On the resulting event, the true counterpart uniquely minimizes $T^{(n)}(v, u)$ for every v in every seed layer, so PROFILESEEDS returns exactly the true pairs in those layers.

Finally,

$$\sum_{j \in \mathcal{J}_{\text{seed}}} N_j \asymp n \sum_{j \geq k_{\max} - \alpha(\beta-1)^{-1} \log_2 n} 2^{-(\beta-1)j} = n^{\alpha+o(1)}, \quad (84)$$

so the seed set has polynomial size. $\square$

C Proof of full witness propagation

Proof of Proposition 4.2. Assume L_j is fully and correctly matched. For a true pair (v, v) with $v \in L_\ell$, the witness count satisfies

$$A_j^{(n)}(v, v) \sim \text{Bin}(N_j, s^2 p_{\ell j}), \quad \mu_{\text{true}} = s^2 N_j p_{\ell j}. \quad (85)$$

By assumption, $\mu_{\text{true}} \geq C \log n$, where C is the constant in the proposition. Chernoff's inequality gives

$$\mathbb{P} \left[A_j^{(n)}(v, v) \leq \mu_{\text{true}}/2 \right] \leq \exp(-\mu_{\text{true}}/8). \quad (86)$$

Since $N_\ell \leq n$, choosing C large enough makes

$$\mathbb{P} \left[\exists v \in L_\ell : A_j^{(n)}(v, v) \leq \mu_{\text{true}}/2 \right] \leq n \exp(-C \log n/8) \leq n^{-20}. \quad (87)$$

For a false pair (v, u) with $u \neq v$, the target layer L_ℓ and witness layer L_j are distinct in the propagation steps, so the potential witness vertices are different from both endpoints. For each $w \in L_j$, the two parent edges (v, w) and (u, w) are independent, and the child-retention variables are independent as well. Therefore

$$A_j^{(n)}(v, u) \preceq \text{Bin}(N_j, C s^2 p_{\ell j}^2), \quad \mu_{\text{false}} \leq C s^2 N_j p_{\ell j}^2 \leq C' p_{\ell j} \mu_{\text{true}}. \quad (88)$$

Since $p_{\ell j} \rightarrow 0$, for large n we have $\mu_{\text{false}} \leq \mu_{\text{true}}/100$. If $X \sim \text{Bin}(N_j, C s^2 p_{\ell j}^2)$, then the Chernoff upper-tail bound gives

$$\mathbb{P} [X \geq \mu_{\text{true}}/2] \leq \left(\frac{e \mu_{\text{false}}}{\mu_{\text{true}}/2} \right)^{\mu_{\text{true}}/2} \leq (C'' p_{\ell j})^{\mu_{\text{true}}/2}. \quad (89)$$

The factor $C'' p_{\ell j}$ tends to zero. Hence, after increasing C , the last display is at most n^{-30} uniformly over all admissible layers. Union bounding over at most n^2 false pairs shows that no false pair reaches the true-pair lower threshold. On the intersection of this event with the true-pair lower-tail event, the true counterpart has witness score at least $\mu_{\text{true}}/2$ and every false candidate has score below $\mu_{\text{true}}/2$, so the true counterpart is the unique maximizer for every $v \in L_\ell$. The failure probability is at most n^{-10} after adjusting constants. $\square$

D Proof of threshold propagation

Proof of Proposition 4.3. Let S_j be the correctly matched subset of L_j , with $|S_j| \geq \lambda_j N_j$. For a true pair (v, v) with $v \in L_{j-1}$,

$$\tilde{A}_j^{(n)}(v, v) \succeq \text{Bin}(\lambda_j N_j, s^2 p_{j-1, j}). \quad (90)$$

By Lemma 2.2, the mean is at least

$$\mu_j \geq C_3 \lambda_j 2^{(3-\beta)j} \quad (91)$$

for some $C_3 = C_3(\beta, s) > 0$. Since $p_{j-1,j} = O(2^{2j}/n)$ and $j = O(\log \log n)$ throughout this stage, $p_{j-1,j} = o(1)$ uniformly. If $X \sim \text{Bin}(m, p)$ with $mp = x$ and $p = o(1)$, then

$$\mathbb{P}[X \geq 3] = 1 - (1-p)^m - mp(1-p)^{m-1} - \binom{m}{2}p^2(1-p)^{m-2} \quad (92)$$

$$= 1 - e^{-x} \left(1 + x + \frac{x^2}{2}\right) + o(1) = g(x) + o(1), \quad (93)$$

uniformly for the values of x arising here. Applying this with $x = \mathbb{E}[\tilde{A}_j^{(n)}(v, v)]$ and using the monotonicity of g gives

$$\mathbb{P}[\tilde{A}_j^{(n)}(v, v) \geq 3] \geq g(C_3 \lambda_j 2^{(3-\beta)j}) - o(1) \geq (1 - o(1))g(C_3 \lambda_j 2^{(3-\beta)j}), \quad (94)$$

where the last step uses the definition of j_{thresh} , which makes the displayed g -term bounded below by $1/2$ during the induction. The indicators for different $v \in L_{j-1}$ depend on disjoint edge sets to S_j . Hence, conditional on S_j , the number of vertices whose true-pair score reaches threshold stochastically dominates a binomial variable with parameters N_{j-1} and the last displayed acceptance probability. Chernoff's lower-tail bound gives at least the claimed number of true scores above threshold, with room for the factor $1 - n^{-1/3}$. On the no-false-pair event proved next, every above-threshold true score is accepted as the true pair by the one-to-one greedy routine, since no false candidate has score at least 3.

For a false pair $(v, u) \in L_{j-1}^{(1)} \times L_{j-1}^{(2)}$, $u \neq v$, the witness count is stochastically dominated by a sum of N_j independent Bernoulli variables whose total mean is at most

$$\nu_j \leq C N_j p_{j-1,j}^2 \asymp C n 2^{-(\beta-1)j} \frac{2^{4j}}{n^2} = C n^{-1} 2^{(5-\beta)j}. \quad (95)$$

Since $j = O(\log \log n)$, $\nu_j = n^{-1+o(1)}$. For any sum X of independent Bernoulli variables with mean ν_j ,

$$\mathbb{P}[X \geq 3] \leq \mathbb{E}\left[\binom{X}{3}\right] \leq \frac{\nu_j^3}{6} = n^{-3+o(1)}. \quad (96)$$

Therefore

$$\mathbb{P}[\tilde{A}_j^{(n)}(v, u) \geq 3] \leq n^{-3+o(1)}. \quad (97)$$

A union bound over at most n^2 false pairs gives no false threshold crossing with probability $1 - n^{-1+o(1)} = 1 - o((\log n)^{-1})$. On this event, every accepted pair is true. $\square$

E Additional comments on assumptions

The dyadic model is chosen for algebraic clarity. The same proof strategy should extend to smoother Chung–Lu power laws if one replaces exact layers by degree bins whose widths are large enough for concentration but small enough that the one-hop witness means remain separated.

The present statement keeps the dyadic grid, known β , and known s to avoid obscuring the core mechanism.

The observed-degree theorem removes the revealed-layer assumption only above logarithmic degree. Below that scale, degree concentration cannot isolate latent layers uniformly over n vertices. This is why the final stage uses global witness thresholding rather than empirical layer reconstruction.

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